Arnav Kewalram

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Education

- **Lynbrook High School** – Rising Senior
- **The Coding School:** Introduction to Artificial Intelligence (2 semesters, sponsored by U.S. Department of Defense)
- **De Anza College:** Summer coursework: Cybersecurity I, II, III

Research Experience

Stanford University — AIMI Lab *June 2026 – Present*
Student Research Intern

NeurIPS Research Paper & Poster *July 2025 – December 2025*
Independent Researcher

- Conducted research on continual learning in LLMs
- Studied interaction between low-bit quantization and replay buffers
- Designed experiments analyzing stability, forgetting, and efficiency trade-offs
- Wrote and presented a poster at NeurIPS 2025 (San Diego)
- Paper: 8-bit Quantization Improves Continual Learning in Large Language Models

ThinkNeuro — Research Intern *July 2023 – December 2023*

- Conducted literature review on psychedelic-assisted treatments
- Co-created research poster on advancements in psychedelic medicine
- Contributed to educational book on mental health
- Participated in community outreach
- Poster: Advancements of Psychedelic Drugs as Treatments

Leadership & Teaching

Schoolhouse.world — AI/ML Coach & Community Moderator *Oct 2024 – Present*

- Teach Intro to AI/ML to middle and high school students
- Design lesson plans and guide hands-on projects
- Support learners across all experience levels

Brighter Future Learning Center — Academic Tutor *June 2024 - August 2024*

- Tutored students in mathematics, delivering 100+ hours of one-on-one instruction

Machine Learning Club — Vice President *May 2025 – Present*

- Organized meetings, workshops, and hackathons
- Coordinated collaborations with other clubs
- Developed and maintaining club website: lhsmlclub.vercel.app

Music Club — Vice President *May 2025 – Present*

- Organized performances, talent shows, and collaborative events

Lynbrook Tech Team *June 2025 – Present*

- Manage school Job Shadow website
- Coordinate connections between recruiters and students

National Honor Society *Sept 2025 – Present*

- Tutor students in Precalculus, Chemistry, Biology, Physics, Algebra, Algebra II/Trigonometry

Independent Projects

AntivirusGAN

- Designed malware detection system using adversarial training
- Red Team generates evolving malware samples
- Blue Team detects threats using behavioral, memory, and network data
- Implemented sandboxed execution and custom evaluation metrics using PyTorch

Music Transcription System

- Built deep learning system converting raw audio into sheet music
- Implemented custom neural networks with Python/PyTorch
- Integrated FFT and librosa for signal processing
- Supported polyphonic audio, live microphone input, and batch processing
- Achieved 87% note detection accuracy

Stock Prediction System

- Built forecasting models: Transformers, LSTM+Attention, CNN-LSTM hybrids, TCNs
- Explored Neural ODEs, Neural Processes, and meta-learning
- Conducted sentiment analysis, anomaly detection, market regime classification

- Implemented uncertainty estimation and ensembles

Recursive Agentic Tree System

- Recursive multi-agent architecture for task decomposition and validation
- Executor agents solve subtasks, checker agents validate outputs
- Focused on interpretability, failure detection, iterative refinement

SwimCoach

- iOS app for real-time swim stroke classification using a custom Temporal Convolutional Network trained from scratch in PyTorch
- Exported model to CoreML and deployed on-device with Swift; built full training pipeline with weakly-supervised video labels

Systems, Robotics, & Engineering

Homelab / NAS (In Progress)

- Building personal NAS and homelab environment
- Focus on storage, virtualization, networking, reliability

PC Building

- Built custom PC from components
- Gained experience in hardware compatibility and troubleshooting

VEX Robotics Team 1439P

- Programmer responsible for autonomous routines and motion control
- Implemented Monte Carlo localization for real-time position estimation

DIY Electronics

- Built electronics projects: electronic car, radio, piano, Arduino systems

Cybersecurity Education Initiative

Founding Member 2025 – Present

- Building platform to teach cybersecurity concepts to beginners
- Developing curriculum with hands-on labs and real-world examples
- Pending formal nonprofit registration

Publications & Writing

- Research Paper: When Less is More: 8-bit Quantization Improves Continual Learning in Large Language Models
- Medium article: Malware Detection Using Machine Learning
- Research poster: Advancements of Psychedelic Drugs as Treatments

Skills

- **Programming:** Python, Java, C++, Swift, HTML
- **Machine Learning:** PyTorch, GANs, Transformers, LSTMs, Continual Learning, Replay Buffers, Quantization, LangChain, PEFT, CoreML, bitsandbytes
- **Data & Signal Processing:** FFT, librosa, GIS, spatial analysis
- **Systems & Security:** Linux, Docker, sandboxed execution, malware analysis, networking, virtualization
- **Robotics:** Motion control, Monte Carlo localization

Awards & Activities

- USACO Gold
- French Grand Concours Silver (2×)
- Black Belt, Taekwondo
- Marching Band, Band, Swimming, Math Club, Computer Science Club